

Circuit Lab C Test

OCTOBER 25, 2025

RICKARDS SO INVITATIONAL



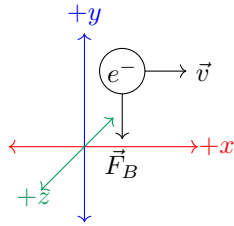
Written By: Shiven Gupta, Singapore Armed Forces '27, & Vikram Nguyen, Pembroke '26

Page:	2	3	4	5	6	7	8	9	10	11	12	13	Total
Points:	10	12	14	16	13	4	10	15	6	28	22	38	188
Score:													

1. The points per question are boxed in the margin next to the question.
2. You may separate the test, but ensure you put it back in order.
3. Out-Of-Browser time will **NOT** be tracked.
4. You may remove materials from your binder **AFTER** the time has started.
5. All answers to be 3 significant figures, and all must be unitless.
6. Tiebreaker questions have **TB#(order)** in front of them. (QN6, QN9, QN21, QN44, QN58)
7. If you have any questions or concerns, please contact me at shiven8gupta@gmail.com. Please also leave your thoughts on this test at [the form linked to this hyperlink after test release](#).

Multiple Choice [100 pts]

1. [2] An electron traveling in the $+\vec{x}$ direction experiences a magnetic force in the $-\vec{y}$ direction. In which direction is the magnetic field directed?



- A. $+\vec{z}$
 B. $-\vec{y}$
 C. $-\vec{z}$
 D. $+\vec{y}$
2. [2] Which of the following is true about brush-less DC (BLDC) motors?
- A. BLDC motors are just DC motors without brushes
 B. They rotate by induced magnetic fields
 C. BLDC motors are inherently more expensive than BDC motors
 D. BLDC motors are only suitable for high-speed applications
3. [2] Identify the below symbol.
- 
- A. Variable Resistor
 B. Potentiometer
 C. Resistor
 D. Inductor
 E. None of the Above
4. [2] What is the average voltage drop across a silicon diode?
- A. 0.7 V
 B. 0.46 V
 C. 1.1 V
 D. 0.23 V
5. [2] Faraday cages are enclosures used to block electromagnetic fields. Which of the following is true about them?
- A. Faraday cages cannot block stable or slowly varying magnetic fields
 B. Near-field, high-powered frequency transmissions like HF RFID are less likely to penetrate
 C. They can block high-frequency electromagnetic pulse (EMP) waves
 D. Mesh cages are nearly as effective as solid cages

6. 2 TB#1 Identify the below symbol.



- A. Signal Ground
- B. Chassis Ground
- C. Neutral Ground
- D. Protective Ground

7. 2 Maxwell-Faraday's Law can be written as (in differential form):

$$\vec{\nabla}_x \times \vec{E} = \frac{\partial \vec{B}}{\partial t}.$$

It states that a time-varying magnetic field always accompanies a spatially varying non-conservative electric field. Which of the following is true about its derivation?

- A. Maxwell mathematically formalized this law in the 1810s
- B. Maxwell mathematically formalized this law in the 1760s
- C. Heaviside helped recast Maxwell's equations into vector calculus form
- D. Descartes helped recast Maxwell's equations into vector calculus form

8. 2 Which of the following are possible switches?

- A. Single-Pole Single-Throw (SPST)
- B. Double-Pole Double-Throw (DPDT)
- C. Single-Pole Double-Throw (SPDT)
- D. Triple-Pole Triple-Throw (TPTT)
- E. All of the Above

9. 2 TB#2 Identify the below symbol.



- A. Full Photodiode
- B. Full Zener Diode
- C. Full Schottky Diode
- D. Full Varicap

10. 2 Which of the following is **false** of MOSFETs?

- A. They are commonly fabricated by controlled silicon oxidation
- B. MOSFETs are a type of buzzer
- C. Physicist Julius Edgar Lilienfeld first proposed the concept of one in 1925
- D. The metal (M stands for metal) is sometimes a misnomer

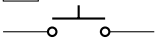
11. 2 What is an iron nail's electromagnetic field strength not dependent on?

- A. Number of wire coils (N)
- B. Number of wire coils per unit length ($n = N/L$)
- C. Supply voltage (V)
- D. Vacuum permeability (μ_0)

12. [2] Who is the SI Unit of Voltage named after, and what does Voltage represent?
- A. Ohm Patel, Voltage represents electron flow.
 - B. John Voltage; Voltage represents electron flow.
 - C. Francois-Marie Arouet Voltaire; Voltage represents potential difference.
 - D. Alessandro Giuseppe Antonio Anastasio Volta; Voltage represents potential difference.
13. [2] Which direction do magnetic fields always point in?
- A. North to South
 - B. South to North
 - C. Left to Right
 - D. Right to Left
14. [2] What type of generator is most commonly used in applications like modern wind turbines?
- A. Permanent magnet
 - B. DC
 - C. Three- ϕ AC
 - D. Tesla Coil
15. [2] What does the "Tesla" (T) measure in SI?
- A. Magnetic Field Strength
 - B. Magnetic Force
 - C. Magnetic Flux Density
 - D. Magnetic Energy per Second
16. [2] What is the resistance of the below resistor?
(In case of printing errors: violet, red, blue, black, green, gray)

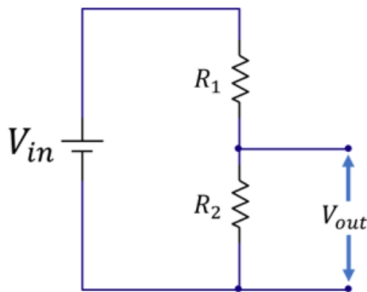


- A. $736 \Omega \pm 0.5\%$
 - B. $736 \Omega \pm 0.5\%$, $1 \text{ ppm}/K$
 - C. $736 \Omega \pm 1\%$, $2 \text{ ppm}/K$
 - D. $638 \Omega \pm 0.5\%$, $1 \text{ ppm}/K$
17. [2] Which of the following is the least common material used to make printed circuit boards (PCBs)?
- A. Kapton
 - B. Aluminum
 - C. Woven Glass
 - D. Phenolic Paper
18. [2] Which of the following give the average difference between LED maximum forward voltages and typical forward voltages?
- A. $0.4 - 0.8 \text{ V}$
 - B. $0.7 - 0.9 \text{ V}$
 - C. $0.2 - 0.3 \text{ V}$
 - D. $0.1 - 0.7 \text{ V}$

19. [2] Which of the following is a true statement on circuits and electromagnetism?
- A. $\sum \Phi_B = \sum \vec{B} \cdot \vec{A} \neq 0$
 - B. $\sum I_{loop} = \frac{\varepsilon}{R}$
 - C. $R_{parallel\ eq} = \frac{1}{\sum_{i=1}^n \frac{1}{R_i}}$
 - D. $\sum V_{loop} = 0$
20. [2] What is a BJT?
- A. Bipolar Junction Transistor
 - B. Bipolar Joint Termination
 - C. Base Junction Transistor
 - D. Binary Junction Transformer
21. [2] **TB#3** A button magnet and a plastic cylinder of the same size are dropped into identical conductive tubular rods. What must be observed?
- A. No difference in fall times is observed
 - B. Inducted currents on the conductive rod speed up the magnet's descent
 - C. The magnet's descent is significantly slower
 - D. The magnet begins to levitate and does not fall all the way through
22. [2] What is true about Joule heating?
- A. Does not occur in superconductive lines
 - B. For DC, $E_{loss} = I(V_1 - V_2)$
 - C. $P = J \cdot E \cdot V = J \cdot E \cdot (IR)$
 - D. Also known as the Joule-Lenz Law in European Union countries
23. [2] Which of the following is not a valid way of calculating electrical current?
- A. $I = nAv_d e$
 - B. $I = ie$
 - C. $I = J \cdot A$
 - D. $I = Qt$
24. [2] Which of the following is true about the skin effect?
- A. AC resistance by skin effect can be increased by using litz wire
 - B. It is the tendency for AC to be greatest near conductor surface
 - C. Tubular conductors do not have any advantage over standard lines
 - D. AC current density decreases exponentially deeper into the conductor
25. [2] Identify this circuit symbol.
- 
- A. Reed
 - B. Rotary Switch
 - C. Toggle Switch
 - D. Push Button
26. [2] These are used to store energy in the form of a magnetic field.

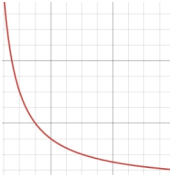
- A. Inductors
- B. Transistors
- C. Capacitors
- D. Batteries

27. [2] High voltage power lines ionize the air around them in a phenomenon known as
- A. The Skin Effect
 - B. The Corona Effect
 - C. Joule Heating
 - D. Capacitive Loss



28. [2] The above circuit can best be described as a
- A. Parallel circuit
 - B. Voltage divider
 - C. Low-pass filter
 - D. Bridge rectifier
29. [3] Based on 3, what is the equation for V_{out} ?
30. [2] Voltaic cells direct electrons from a ____ anode and a ____ cathode, and electrolytic cells direct them from a ____ anode and ____ cathode.
- A. Positive, negative / negative positive
 - B. Positive, negative / positive, negative
 - C. Negative, positive / negative, positive
 - D. Negative, positive / positive, negative
31. [2] This is used to step down AC voltage levels from power lines into houses
- A. Transformer
 - B. Voltage divider
 - C. MOSFET
 - D. Rectifier
32. [2] In an RC circuit, it takes how many time constants for the capacitor to discharge fully?
- A. 1
 - B. 3
 - C. 5
 - D. 7

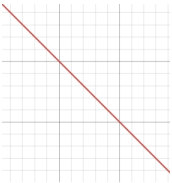
33. ☐ 2 The graph of V vs $n\tau$, where n is an integer, is best modeled by which graph (ignore scale)?



A.

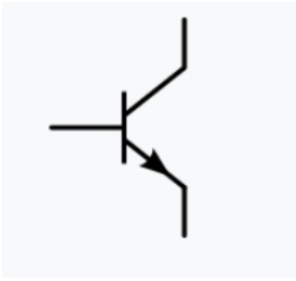


B.



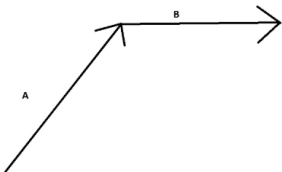
C.

34. ☐ 2 Transistors do NOT
- A. Amplify signals
 - B. Compare voltages
 - C. Absorb excess current
 - D. Act as switches



35. [2] What type of transistor is this?

- A. BJT
- B. NPN
- C. PNP
- D. MOSFET



36. [2] In which way does $\mathbf{A} \times \mathbf{B}$ direct?

- A. Into the screen
- B. Out of the screen
- C. The direction of the resulting vector formed by connecting the tail of $\mathbf{A}$ and the tip of $\mathbf{B}$
- D. None of the above, $\mathbf{A} \times \mathbf{B}$ is a scalar

37. [2] Once charged, capacitors _____ AC and _____ DC

- A. Pass / pass
- B. Pass / block
- C. Block / pass
- D. Block / block

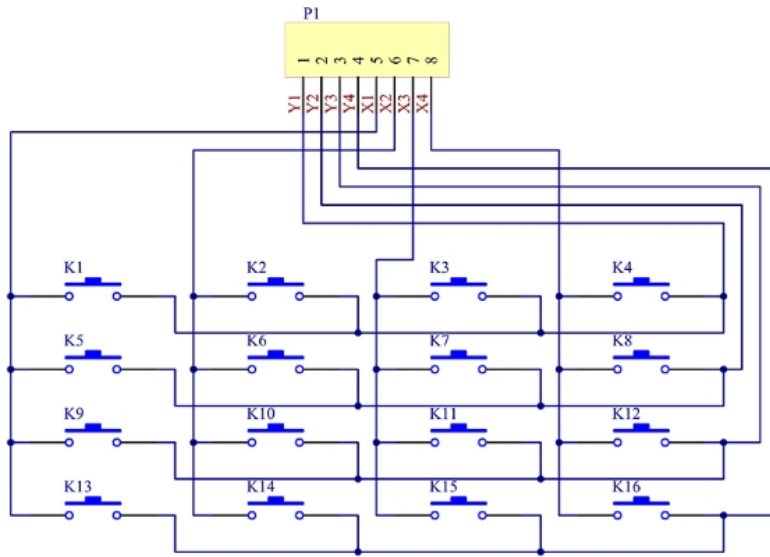
38. [2] What is the most appropriate hydraulic analogy for an inductor?

- A. A tighter pipe
- B. A storage tank
- C. A flywheel
- D. A one-way valve

39. [2] A stream of Na^+ ions flows through a cell membrane in $3.5 \mu\text{s}$. What is the current, I ?

- A. $5.49 \times 10^{-11} \text{ A}$
- B. $4.51 \times 10^5 \text{ A}$
- C. $1.60 \times 10^{-19} \text{ A}$
- D. 1.00 A

40. [2] A hair dryer draws 7.5A from a 120V line. What is the resistance?
- A. 0.063 Ω
 - B. 1.6 Ω
 - C. 16 Ω
 - D. 18 Ω
41. [2] How much charge passes through it in 15 minutes? (Assume DC)
- A. 112.5 C
 - B. 112.5 A
 - C. 7500 C
 - D. 6.7 kC haha mangos
42. [2] Why is AC transmitted at a high voltage when it must be stepped down for usage in household appliances?
- A. Higher voltage means lower current and therefore less power loss
 - B. As voltage increases, joule heating from the Skin Effect decreases to the point of negligibility
 - C. AC is inefficient over long distances because of its constant polarity shifts
 - D. Multiple answers are correct
43. [3] Provide an equation that explains the answer(s) to the previous question.
44. [2] **TB#4** Consider a simple voltage divider with an input voltage of 7.5V, with $R_1 = 5 \Omega$ and $R_2 = 8 \Omega$. What is the output voltage?
- A. 2.88 V
 - B. 4.62 V
 - C. 3.74 V
 - D. 577 mV
45. [2] Why do most microprocessors use DC instead of AC?
- A. Semiconductors cannot function at alternating frequencies
 - B. Logic gates require binary signals
 - C. Capacitors in electronics block AC
 - D. Pullup resistors in most microprocessors cannot handle the repeated change in polarity that AC induces
46. [2] Calculate the electrostatic force between 2 electrons 200 picometers apart.
- A. $3.2 \times 10^{-10} \text{ N}$
 - B. $5.8 \times 10^{-9} \text{ N}$
 - C. 9.0×10^{-29}
 - D. $3.6 \times 10^{-10} \text{ N}$



47. ☒ 2 The device shown by the schematic above is most likely a
- Matrix keypad
 - Crossbar switch
 - Resistor network / resistor bank
 - LED display driver
 - There is not enough information to determine the device
48. ☒ 2 A fuse may be preferential to a breaker because it does NOT
- Respond faster
 - Have resuability potential
 - Pose a touching hazard
 - Cost as much
 - None of the above
49. ☒ 2 Which is true about two wires of the same material but different dimensions?
- The longer wire has higher resistivity
 - The thicker wire has higher resistivity
 - Both wires have the same resistivity
 - There is not enough information to answer
50. ☐ 0 Which Clash Royale card can NOT be uncharged by means of electricity?
- Tesla
 - Sparky
 - Inferno Dragon
 - Zappies
 - Mighty Miner

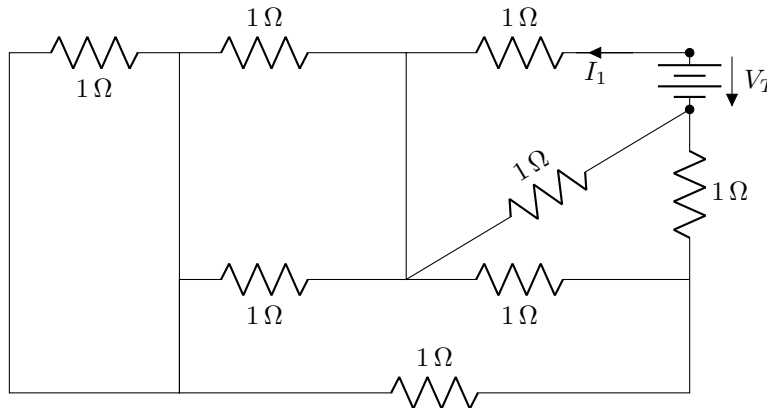
Free Response [90 pts]

51. [2] What average speed does energy flux Φ_E travel through a circuit?

52. [2] What average speed do electrons in a circuit flow at?

53. [4] The complex impedance of a circuit is given by $Z = 67i^2 + 21 - 42i (\Omega)$. What is the absolute impedance $|Z|$? Give your answer rounded up to 2 decimal places.

54. [9] Resolve the equivalent resistance of the below circuit. Take each resistor $R = 1 \Omega$.



55. [3] For an unideal wire of $\rho = 5 \Omega/km$, $\ell = 5.00 km$, and cross-sectional area of $A = 0.0025 km^2$, what is the resistance R ?

56. [3] Approximately how many time constants τ_{LR} are needed for an LR circuit to reach its steady-state current?

57. [3] Approximately how many time constants τ_{RC} are needed for a capacitor in an RC circuit to reach its full charge?

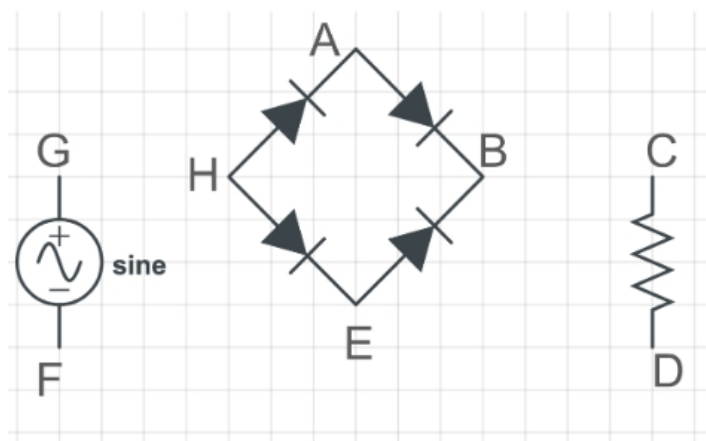
58. [2] **TB#5** The effective resistance of a circuit is 15Ω . What is the real component of the impedance $Z_{re} \in \mathbb{R}$?

59. [4] What is the charge flowing across a wire per second for a wire of diameter 0.0001 m , electron drift speed of $1 \times 10^{-5}\text{ m/s}$ and a electron number density of $1 \times 10^{28}\text{ /m}^3$?

60. [3] What is the resistivity of a conductor with current density 5.00 A/m^2 and electric field strength of 2.00 V/m ?

61. [6] What is the magnetic flux through a conductor with potential difference $\epsilon\text{ (V)}$, length $\ell\text{ (m)}$, resistivity $\rho\text{ (}\Omega\text{m)}$, radius $r\text{ (m)}$? Assume that it is also connected to a capacitor in the circuit, and that the effective capacitance is $C\text{ (F)}$ and the resonant frequency is $\omega\text{ (rad/s)}$. Also, state the unit of measurement.

62. [2] Explain the difference between full-wave and half-wave rectifiers.



63. [4] Draw lines and connect the parts to create a circuit that forces a unidirectional flow of current. If taking digitally, list the connections as line segments (ie AB, BC, CD...)

64. [3] Smooth the output voltage of the above circuit by adding a single component. If taking digitally, identify the component and describe where to put it.

65. [6] Consider a magnetic field of 0.02T directed orthogonally through a single coil with a diameter of 50cm. Calculate the magnetic flux.

66. [8] Assuming a transformer with $N_1 = 200$ and $N_2 = 800$ reaches this value from 0 in a time interval 0.02s, calculate the average EMF in both coils.

67. [6] Explain how transformers utilize magnetic flux to change the voltage AND whether this one is stepped up or down.

68. [4] A circuit viewed from above carries current clockwise in a single loop/coil. Determine the direction of the magnetic field from the center of the circuit. Does this direction change when the current switches direction? If so, which direction? If not, explain why.

69. [2] Does this direction change when the current moves in a straight line from left to right instead of a loop? If so, which direction? If not, explain why.



70. [2] Identify the resistance. In case of printing errors: orange, blue, brown, gold

71. [4] A 9V battery and a red LED are connected in series with this resistor. Calculate the current. Is this reasonable for an LED?

72. [4] Suppose a capacitor of 100 microfarads were placed parallel to the LED. What is the RC time constant, and how long would it take to fully discharge once the battery is removed?

73. [2] It's generally said that "current kills, not voltage", as voltage is simply the driving force behind current; current as small as 0.2A is enough to be lethal for humans. Why, then, do warning signs around large power sources say "high voltage"?

74. ☐ Find the directional derivative of $z = f(x, y) = e^x(\sin(y))$ at $(0, 3)$ in the direction of $v = \langle -6, 8 \rangle$ (hint: $\nabla z = \langle \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y} \rangle$) Then, email my math teacher and let him know that I am geeked for the test this week.